A Brief Introduction to Combinatorial Reconfiguration and Related Problems

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Short Bio - Hoàng Anh Đức

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- **Education:**
  - **B.Math** @ VNU-HUS, Vietnam (2008–2013).
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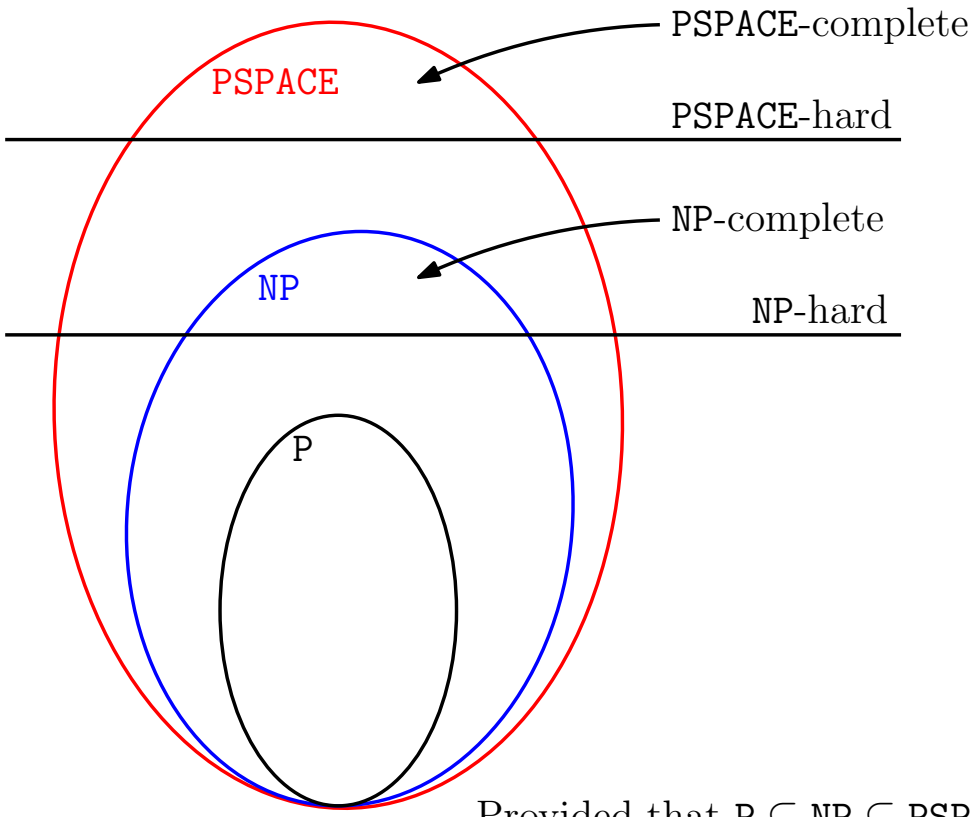
Introduction

Remind

- We always talk about *decision problems* (output YES or NO)
- Complexity Classes
  - P: Problems can be "solved efficiently" in polynomial time
  - NP: Problems can be "verified efficiently" in polynomial time
  - PSPACE: Problems can be "solved efficiently" in polynomial space



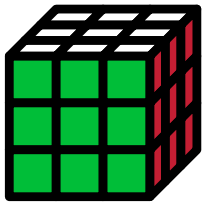
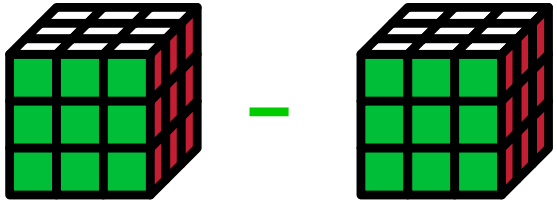
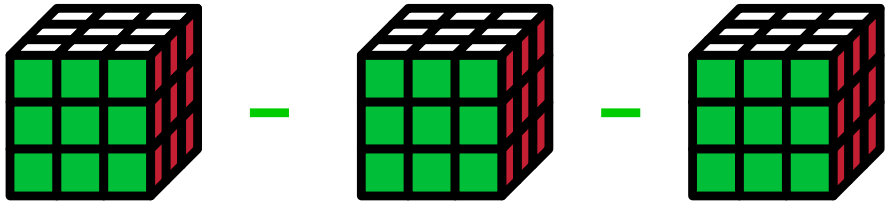
Provided that $\mathtt{P} \subsetneq \mathtt{NP} \subsetneq \mathtt{PSPACE}$

Definition

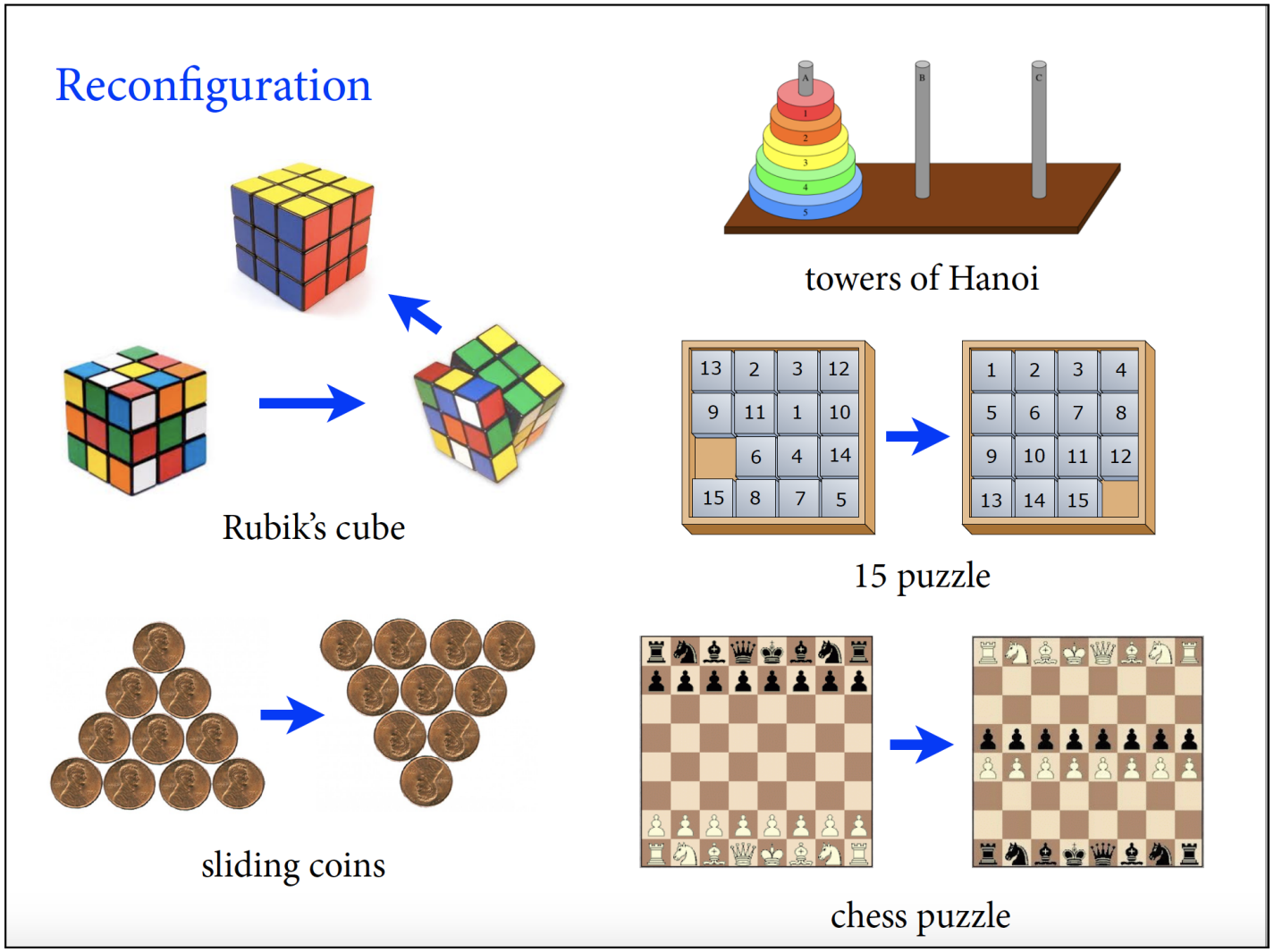
- **Input:**
  - description of what *states* ($\equiv$ configurations) are
  - *allowed moves* between states ($\equiv$ reconfiguration rule)
- **Question:**
  - Reachability: Given two states S and T, is there a sequence of moves that *transforms S into T*?
  - Shortest Transformation: Given two states S and T and some positive integer ℓ, is there a sequence of moves that *transforms S into T using at most ℓ moves*?
  - Connectivity: Is there a sequence of moves *between any pair of states*?
  - Diameter: Is the *maximum distance* between any two states bounded? (distance $\equiv$ smallest number of moves.)
  - and so on

Definition

- Two viewpoints: as a *process* or as a *graph*

Reconf. Process	**Reconf. Graph**	**Example** RUBIK'S CUBE
state	node	
move *[reconf. rule]*	edge	
seq. of moves *[reconf. sequence]*	path	

Reconfiguration games/puzzles



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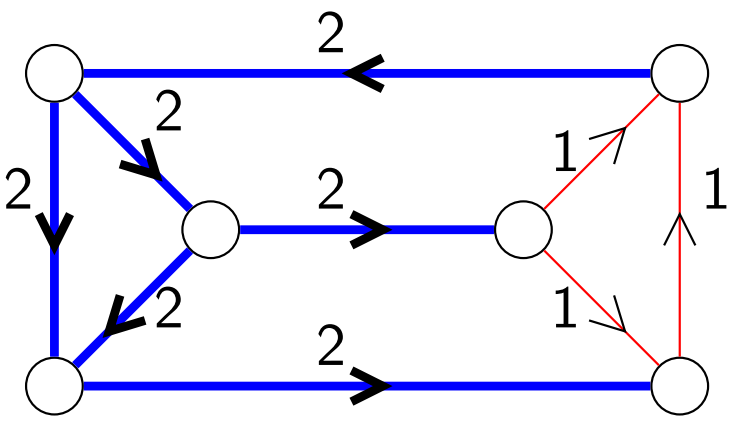
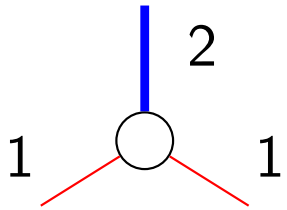
Reconfiguration games/puzzles

Nondeterministic Constraint Logic (NCL) [Hearn and Demaine 2005]

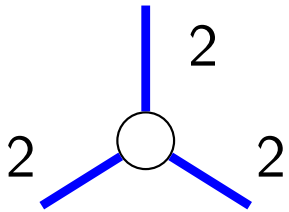
- **Input:**
  - Each *configuration* involves a graph having red (weight 1) and blue (weight 2) edges where each edge is oriented such that *(*) the sum of weights of in-coming arcs at each vertex is at least 2*
  - Each *move* involves re-orienting an edge such that (*) is satisfied
- **Question:** Is there a sequence of moves that transforms one given configuration into another? (PSPACE-complete in general)



(a)



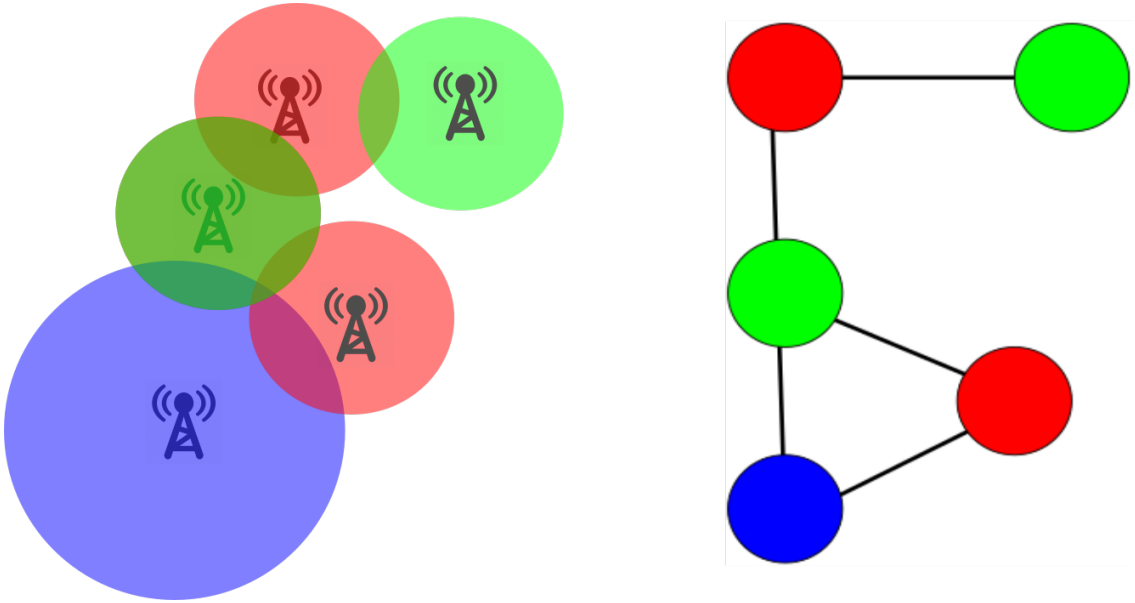
(b)



(c)

Frequency Re-Assignment

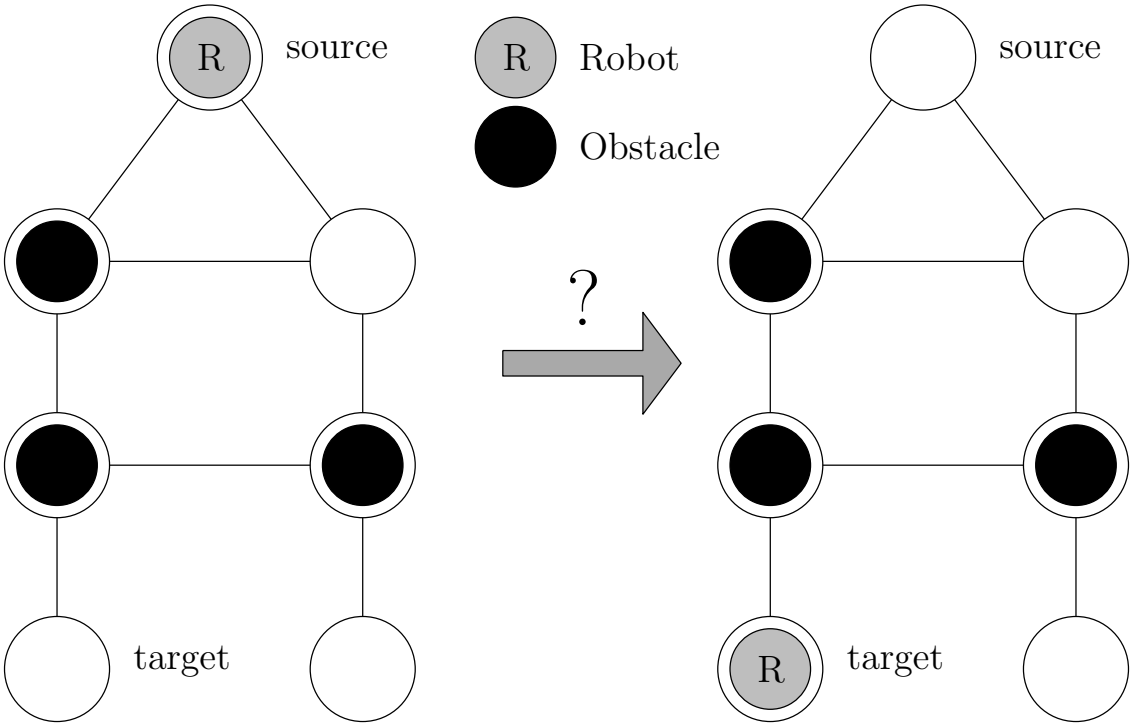
- (*) Two cell towers whose covering areas intersect must have different frequencies.
- Can we reassign the frequency of one tower at a time without affecting (*)?



Frequency Re-Assignment $\equiv$ Vertex-Coloring Reconfiguration

Robot Motion

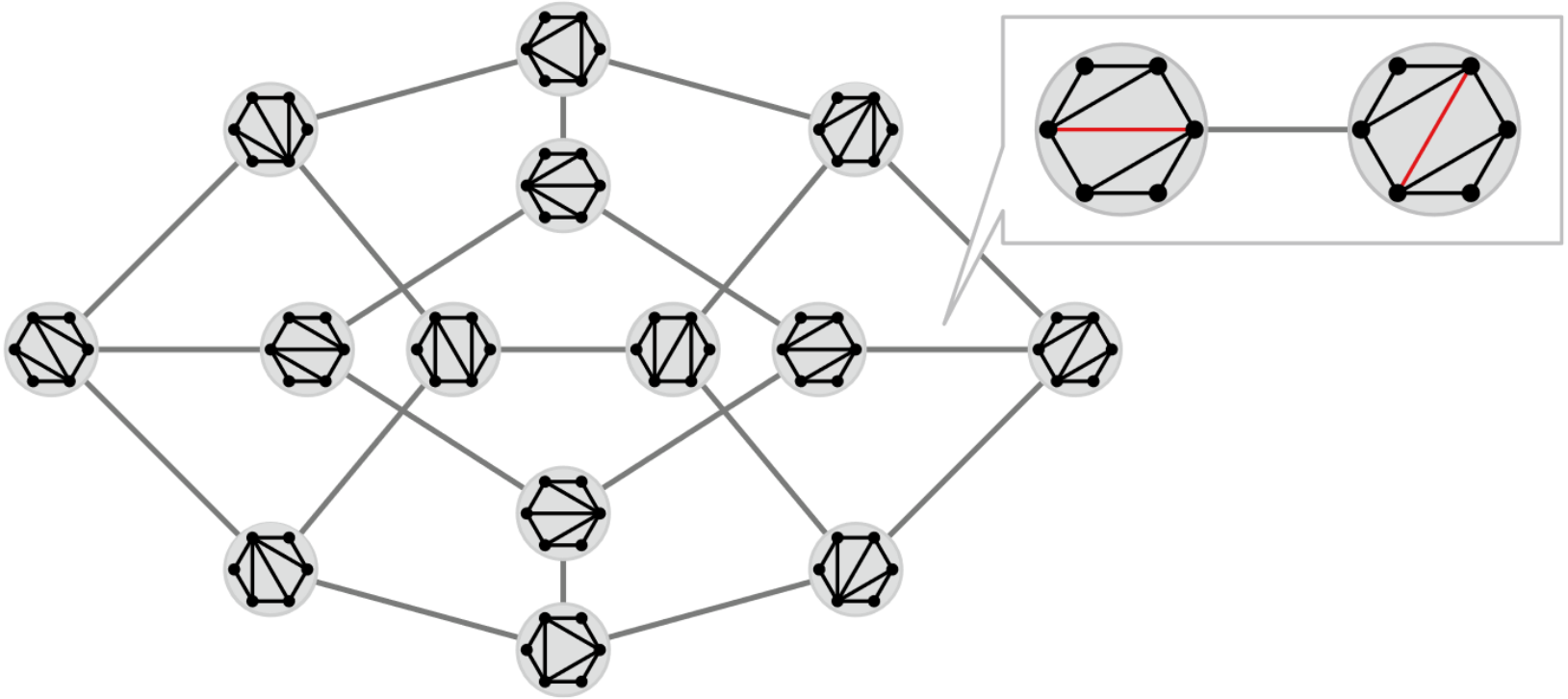
- Robots and obstacles are placed in an environment
- All robots are controlled by a central algorithm to perform some tasks (e.g., managing a warehouse or inventory) as a team
- Can we move robots to their final destinations without having collision with other robots or obstacles?



Robot Motion $\equiv$ Token Reconfiguration [Hopcroft et al. 1984]

Flip Distance Between Triangulations

- **Input:** Two different triangulations S and T of a convex polygon
- **Question:** Is there a sequence of flips that transforms S into T using at most ℓ flips?



Flip graph ($\equiv$ Reconfiguration graph) of triangulations of a convex polygon [Aichholzer et al. 2021]

Extra Information and Resources

Reconfiguration has emerged in different areas, including

- Robot motion [Gupta, Sa'ar, et al. 2020]
- Rerouting shortest paths [Gajjar et al. 2022]
- Knot theory [Russell and Sazdanovic 2021]
- Quantum complexity theory [Gharibian and Rudolph 2022]
- Distributed algorithms [Gupta, Kumar, et al. 2022]
- and so on

Extra Information and Resources

- Surveys in Reconfiguration
  - J. van den Heuvel (2013). "The Complexity of Change". In: *Surveys in Combinatorics*. Vol. 409. London Mathematical Society Lecture Note Series. Cambridge University Press, pp. 127–160. DOI: 10.1017/cbo9781139506748.005
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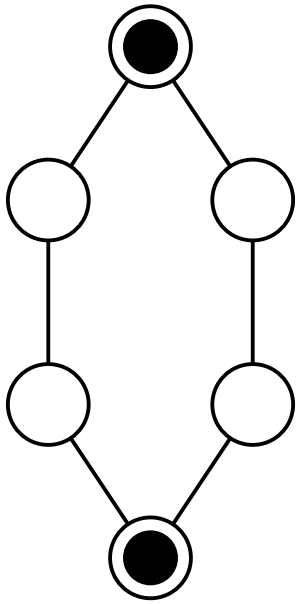
  - N. Bousquet et al. (2022). "A survey on the parameterized complexity of the independent set and (connected) dominating set reconfiguration problems". In: *arXiv preprint*. arXiv: 2204.10526
- Online Wiki: `http://reconf.wikidot.com/`
- "Invitation to Combinatorial Reconfiguration" – Invited Talk by Takehiro Ito at WALCOM 2022.

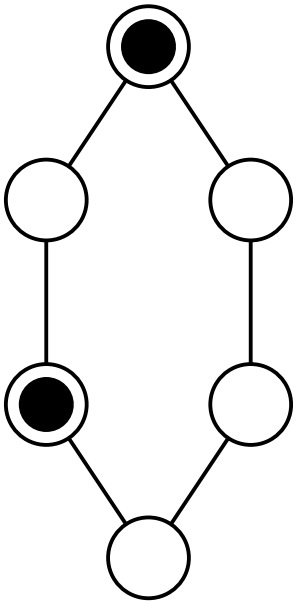
On The Complexity of Distance-d Independent Set Reconfiguration

Introduction

For a fixed integer $d \geq 2$, a *distance-d independent set (DdIS)* of a graph G is a vertex subset where *any two members have distance at least d*. A D2IS is nothing but an independent set (stable set) of G.



D2IS: YES
D3IS: YES



D2IS: YES
D3IS: NO

Introduction

- A DdIS of $G \equiv$ A set of tokens placed on vertices of G where the distance between any two tokens is at least d.
- Two well-known reconfiguration rules:
  - **Token Sliding (**TS**):** a token can only move to one of its neighbors.
  - **Token Jumping (**TJ**):** a token can move to any unoccupied vertex.
- **Distance-d Independent Set Reconfiguration (DdISR) under** $\mathsf{R} \in \{\mathsf{TS}, \mathsf{TJ}\}$
  - **Input:** Two DdISs (token-sets) I, J of G and the reconfiguration rule R.
  - **(Reachability) Question:** Is there a sequence of DdISs that transforms I into J where each intermediate member is obtained from its predecessor by a single R-move?

Our Motivation

D2ISR under TS or TJ are two of the most well-studied reconfiguration problems [Nishimura 2018].

Table 1. Results on reachability for INDEPENDENT SET RECONFIGURATION; question marks indicate problems for which there are no known complexity results.

Class	Source	TS	TAR/TJ
planar (of degree 3)	NP-complete	PSPACE-complete [17]	PSPACE-complete [17]
even-hole-free	?	?	P [17,28,54]
perfect	P	PSPACE-complete [17]	PSPACE-complete [17]
cograph	P	P [17]	P [42,55]
chordal	P	?	P (since even-hole-free)
split	P	co-NP-hard [56]	P (since even-hole-free)
interval	P	P [56]	P (since even-hole-free)
claw-free	P	P [57]	P [57]
proper interval	P	P [58]	P (since even-hole-free)
caterpillar	P	P [58]	P (since even-hole-free)
tree	P	P [59]	P (since even-hole-free)
bipartite	P	PSPACE-complete [52]	NP-complete [52]
bipartite permutation	P	P [60]	?
bipartite distance hereditary	P	P [60]	?
block	P	P [61,62]	?
bounded treewidth	P	PSPACE-complete [47]	PSPACE-complete [47]
bounded pathwidth	P	PSPACE-complete [47]	PSPACE-complete [47]
bounded bandwidth	P	PSPACE-complete [47]	PSPACE-complete [47]
cactus	P	P [61,63]	P [54]

It is natural to consider the computational complexity of DdISR for fixed $d \geq 3$.

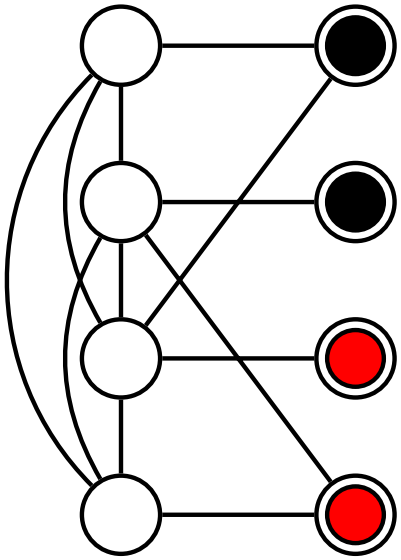
Our Results

- On chordal graphs and split graphs, there are some interesting complexity dichotomies.
  - Under TJ on chordal graphs, DdISR is in P for even d and PSPACE-complete for odd d.
  - On split graphs, DdISR under TS is PSPACE-complete for $d = 2$ [Belmonte et al. 2021] but in P for $d = 3$.
    Under TJ, it is in P for $d = 2$ [Kamiński et al. 2012] but PSPACE-complete for $d = 3$.
- Several known results for $d = 2$ can be extended for $d \geq 3$. More precisely, DdISR ($d \geq 3$) is PSPACE-complete on
  - general graphs, extending the result for $d = 2$ in [Ito et al. 2011]
  - planar graphs of maximum degree 3 and bounded bandwidth, extending the result for $d = 2$ in [Hearn and Demaine 2005]
  - perfect graphs, extending the result for $d = 2$ in [Kamiński et al. 2012]

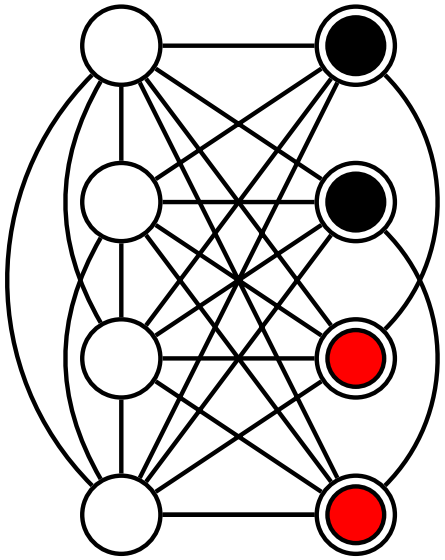
Observations

- Let G be a graph. The *sth power of G* for some $s \geq 1$, denoted by G^s, is the graph having $V(G^s) = V(G)$ and for $u, v \in V(G^s)$, we have *$uv \in E(G^s)$ if the distance between u and v is at most s in G*.
- I is a DdIS of $G \Leftrightarrow I$ is an independent set of G^{d-1}.
- Consequently, let $\mathscr{G}$ and $\mathscr{H}$ be two graph classes and suppose that for every $G \in \mathscr{G}$ we have $G^{d-1} \in \mathscr{H}$ for some fixed integer $d \geq 2$. *If ISR under* TJ *on $\mathscr{H}$ can be solved in polynomial time, so does DdISR under* TJ *on $\mathscr{G}$.*
- However, because G^{d-1} has much more edges than G, *this does not hold under* TS.

Observations



$(G, I, J, \mathsf{TS}, 3)$: NO



$(G^2, I, J, \mathsf{TS}, 2)$: YES

A $\mathrm{D}d\mathrm{ISR}$'s no-instance $(G, I, J, \mathsf{TS}, 3)$ whose corresponding ISR's instance $(G^2, I, J, \mathsf{TS}, 2)$ is a yes-instance. Tokens in I (resp., J) are marked with black (resp., red) color.

Chordal Graphs and Split Graphs

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